

**Overall model comparison for WEATHER**  
**Lookback = 72 steps, Forecast horizon = 12 steps**

| Model         | RMSE   | MAE    | sMAPE (%) | R <sup>2</sup> | Bias   | NRMSE |
|---------------|--------|--------|-----------|----------------|--------|-------|
| CNN           | 33.550 | 10.111 | 64.700    | 0.989          | -0.003 | 0.182 |
| LSTM          | 34.689 | 9.721  | 65.074    | 0.989          | 1.009  | 0.188 |
| dCeNN+ELM     | 38.054 | 12.339 | 69.389    | 0.986          | -0.374 | 0.206 |
| dCeNN+ELM+ASP | 37.123 | 10.742 | 45.127    | 0.987          | -0.007 | 0.201 |

- This table compares all models for one fixed experimental setting.
- For RMSE, MAE, sMAPE, and NRMSE, smaller values are better. For R<sup>2</sup>, larger values are better. For Bias, values closer to zero are better.
- Weather combines multiple physical variables. Aggregate RMSE or MAE should therefore be read mainly as a model-comparison tool, not as a single physical-unit error.